Adaptive preprocessing for TimesFM 3.0 and Chronos-2

What changes when we preprocess before zero-shot forecasting

UConn Senior Design Project · Team 46

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OVERVIEW

Setup and bottom line

- **Data:** 21 series types: 17 synthetic (5 series each) + 4 real datasets with rolling windows, 103 forecasts.
- **Models:** TimesFM 3.0 and Chronos-2, zero-shot, default settings, no covariates.
- **Metrics:** MASE (error ÷ typical step-to-step change; lower is better) and 80%-band coverage (target 80%).
- **Comparison:** raw input vs. adaptive preprocessing, scored on the same held-out windows.

-1.1%

TimesFM 3.0, overall MASE

Geometric mean over all 103 forecasts.

-1.7%

Chronos-2, overall MASE

Most series pass through unchanged.

-21% / -40%

Exponential growth

TimesFM / Chronos-2, from Box-Cox (log). The one big win.

+8% / +29%

Air passengers (worse)

Box-Cox passed the gate but hurt on test.

Why so little change? Both models already normalize, detrend, interpolate gaps and ignore isolated spikes internally.

METHOD

How preprocessing is chosen

If the history shows…	then apply…	TimesFM	Chronos-2
NaN gaps	STL seasonal gap fill	yes	no*
Positive + multiplicative seasonality or exponential growth	Box-Cox (Guerrero λ, often log)	yes	yes
Unit root (ADF p > 0.1) and weak seasonality	Differencing (forecast changes, re-integrate)	yes	yes
Sharp step change, stationary residual	Level-shift alignment	yes	yes
Outliers: 0-3% of points with robust |z| > 7	Outlier repair (STL fit)	yes	yes
Non-negative history	Clip forecast at 0	yes	yes

* Chronos-2 masks NaNs natively, so imputing would feed it invented values. TimesFM fills gaps with straight lines internally.

Gate: keep a recipe only if it earns it

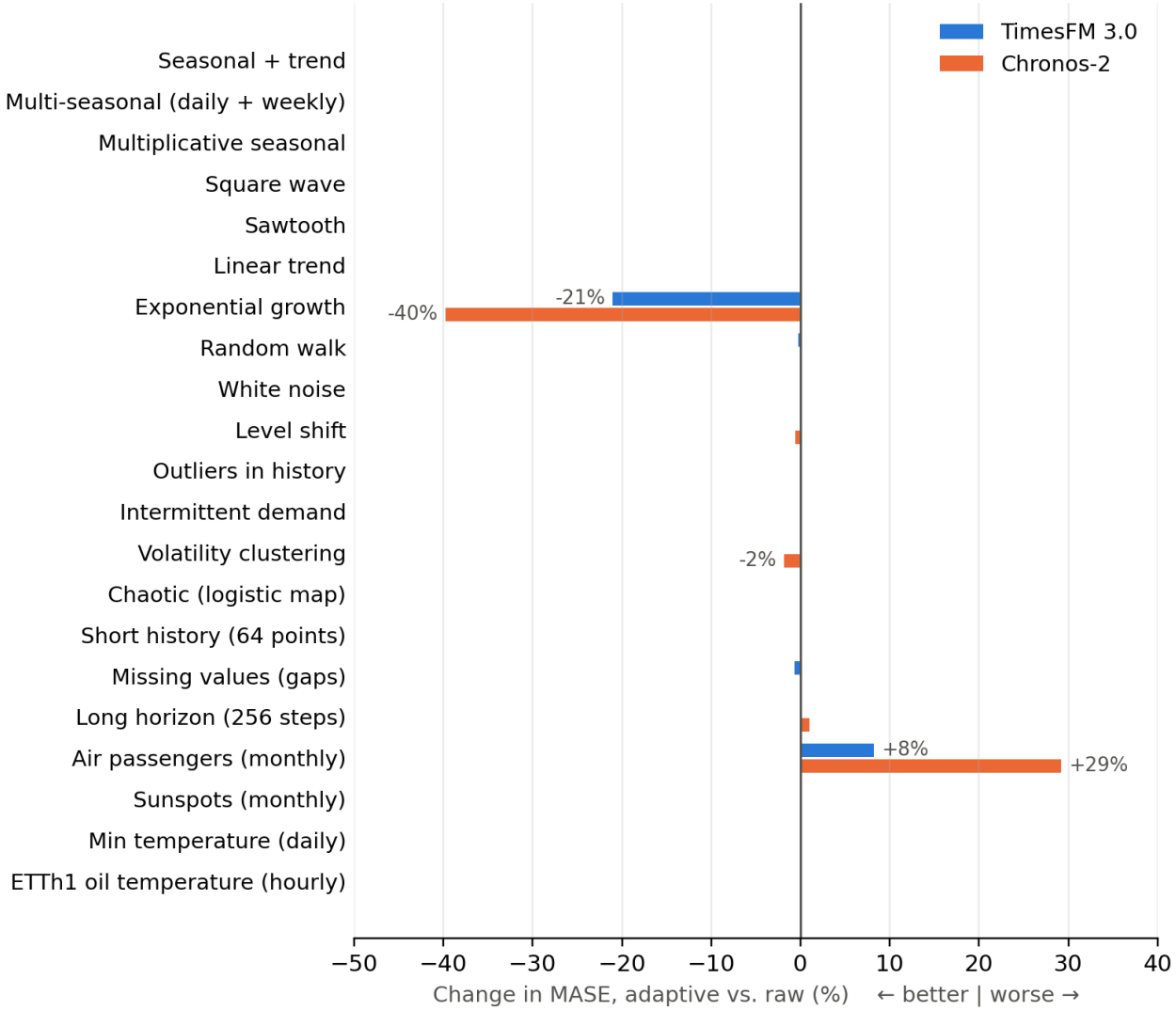
Back-test on the last 3 horizons of the history (no peeking at the test window). Use the recipe only if validation MASE is ≥5% lower than raw AND it wins ≥2 of 3 windows. Otherwise forecast raw.

Calibration: widen-only

If the 10-90% band covered < 70% of validation points, widen it around the median until it reaches 80%. Never narrow: three short windows are too noisy to justify shrinking.

RESULTS

MASE change across all 21 types



- **Exponential growth:** -21% / -40%. Both models are affine-equivariant, so rescaling can't remove compounding growth; a log transform turns it linear.
- **Air passengers:** +8% / +29%. Only 108-132 monthly points, so 12-month validation windows let a bad Box-Cox through.
- **Everything else:** within ±2%. Most bars are empty because the gate kept raw input.

RESULTS

What actually changed

Data type	Steps accepted: TimesFM	Steps accepted: Chronos-2	TimesFM Δ	Chronos-2 Δ
Exponential growth	Box-Cox 5/5, outlier repair 2/5	Box-Cox 5/5	-21.0%	-39.8%
Volatility clustering	differencing 1/5	differencing 3/5, outlier repair 1/5	-0.1%	-1.8%
Missing values (gaps)	STL gap fill 1/5	none (kept raw)	-0.7%	+0.0%
Level shift	none (kept raw)	level-shift align 3/5, outlier repair 1/5	+0.0%	-0.6%
Random walk	differencing 3/5	none (kept raw)	-0.3%	+0.0%
Long horizon (256 steps)	Box-Cox 1/5	Box-Cox 1/5	+0.1%	+1.0%
Air passengers (monthly)	Box-Cox 3/3, outlier repair 2/3	Box-Cox 2/3	+8.3%	+29.2%

Counts = series (out of 5, or 3 for Air passengers) where the gate accepted the step. The other 14 types: MASE unchanged for both models (gate kept raw input, or the accepted step had no effect).

- **Box-Cox** is the only step with a large effect, in both directions.
- **Differencing and level-shift alignment** give small, consistent gains on random-walk, volatility and step-change series.

RESULTS

Uncertainty: 80%-band coverage

	TimesFM before	TimesFM after	Chronos-2 before	Chronos-2 after
Mean over all types	78.5%	79.4%	75.2%	77.8%
Exponential growth	64%	67%	57%	87%
Square wave	76%	78%	67%	84%
Sawtooth	68%	75%	69%	72%
Sunspots (monthly)	74%	79%	72%	82%
Volatility clustering	68%	65%	69%	64%
Air passengers (monthly)	61%	69%	72%	50%

Red = more than 10 points from the 80% target (darker = got worse). Amber = still off, but improved. Green = improved and within target (darker = bigger gain).

- **Fixed:** several too-narrow bands move to target (Chronos-2 square wave 67% → 84%, exponential growth 57% → 87%).
- **Not fixed:** volatility clustering. Differencing sums step uncertainty as if steps were independent, which under-covers when volatility clusters. Air passengers inherits the bad Box-Cox.

CONCLUSION

Takeaways

- **Most preprocessing is redundant** for these models: scaling, detrending, gap filling and outlier repair are already handled internally.
- **Use Box-Cox for exponential or multiplicative growth:** the one consistent, large gain (-21% / -40% MASE).
- **Use differencing for unit-root series:** small gains, and it removes spurious drift.
- **Always gate preprocessing on a back-test,** and use more than three windows on short series (Air passengers shows why).